Name: _____ **Period:** _____ **Date:** _____

Topic 3.4 Guided Notes: Cognitive Development Across the Lifespan

Unit 3: Development and Learning | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic3.4

Word Bank

Theory of Mind	*Assimilation*	*Animism*
Preoperational Stage	*Schema*	*Mental Symbols*
Sensorimotor Stage	*Object Permanence*	*Accommodation*
Pretend Play	*Reversibility*	*Egocentrism*
Conservation		

1. Building Blocks of Thought: Schemas

According to Piaget, our brains are constantly trying to make sense of our experiences. We do this by building a _____—a mental framework or file folder that helps us organize and interpret information. When we encounter something new, we process it using two continuous mechanisms:

- _____: Integrating new information into our existing schemas without changing them. (e.g., A toddler who has a schema for "dog" sees a horse for the first time and calls it a "big dog").
- _____: Changing and modifying our existing schemas to fit new information. (e.g., The toddler learns that horses are a different category and creates a brand-new schema for them).

Apply It: Assimilation or Accommodation?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A child who has always called every four-legged furry pet a "kitty" now confidently calls a new puppy a "kitty" too, without adjusting his thinking at all.	
2	After being corrected several times, a child who initially called all round fruits "apples" learns that oranges are a completely different category and builds a brand-new mental category just for them.	

2. Piaget's Stages of Cognitive Development

Piaget believed that cognitive development happens discontinuously in four distinct stages:

Stage 1: Sensorimotor Stage (Birth – 2 years)

In the _____, infants learn about the world purely through their physical senses and motor actions (touching, looking, grasping). The major milestone they must achieve here is _____—the understanding that objects continue to exist even when they cannot be seen. This is exactly why the game peek-a-boo is so thrilling to a young baby; before they master this milestone, they genuinely believe your face has vanished from existence when you hide it behind your hands!

Stage 2: Preoperational Stage (2 – 7 years)

In the _____, children begin using _____—words, images, and other internal representations that stand in for real objects and ideas. They engage heavily in _____ (such as imagining a cardboard box is a rocket ship) and often exhibit _____, believing inanimate objects have feelings and intentions (such as apologizing to a dropped teddy bear because they think it is sad). However, they lack logical reasoning. Their thinking is characterized by:

- _____: The inability to see a situation from another person's point of view. They assume everyone sees, hears, and feels exactly what they do. This is why a 4-year-old might eagerly wrap up their own favorite toy truck to give to their mom for her birthday—they assume she will love it just as much as they do!
- A lack of _____: They do not understand that quantity remains the same even if the appearance changes. If you pour identical amounts of juice into a short, wide glass and a tall, skinny glass, a preoperational child will confidently claim the tall glass has more juice simply because the liquid line is higher.
- A lack of _____: They struggle to mentally reverse a sequence of events. They might cry if their hamburger is served with pickles, not realizing they can simply take the pickles off to return the burger to its original state.

Toward the end of this stage, children begin to overcome egocentrism by developing a _____—the crucial realization that other people have their own unique thoughts, beliefs, and perspectives that might differ entirely from their own.

Apply It: Which Preoperational Trait?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A little girl insists on saying “sorry” to her stuffed rabbit after accidentally dropping it on the floor, convinced that it felt hurt by the fall.	
2	A boy transforms his living room couch cushions into a pirate ship, complete with an imaginary crew and treasure map, fully immersed in the make-believe scenario.	
3	A little boy refuses to eat his sandwich after his mom cuts it in half, crying that it's now “broken” and wanting a new, whole sandwich instead of realizing the two halves could simply be put back together.	

#	Scenario	Concept
4	A 7-year-old watches a puppet show where a puppet named Sally hides a marble in a basket and then leaves the room; while Sally is gone, another puppet moves the marble to a box. The child now correctly predicts that Sally will still look for the marble in the basket when she returns, understanding that Sally holds a belief different from what the child knows to be true.	

Stage 3: Concrete Operational Stage (7 – 11 years)

In the _____, children overcome the cognitive flaws of the previous stage. They finally grasp conservation and reversibility, meaning they can easily perform mental math operations (understanding that if 4 + 3 = 7, then 7 - 3 = 4). They can think logically about concrete, real-world events and successfully sort physical objects into complex categories. However, they still struggle heavily with abstract, systematic, or purely hypothetical questions.

Stage 4: Formal Operational Stage (12 years – Adulthood)

In the final stage, the _____, individuals break free from needing tangible objects and gain the capacity for _____ and _____. They can systematically ponder "what if" situations they've never directly experienced ("What would happen to the economy if humans no longer needed sleep?"), debate complex ethical dilemmas, and successfully solve algebraic equations where abstract letters represent unknown numbers. Interestingly, Piaget noted that not all individuals fully achieve this advanced level of thinking.

Apply It: Which Piagetian Stage?

Using the word bank above, identify which stage best matches each scenario.

#	Scenario	Stage
1	A 14-month-old baby who drops a toy behind the couch immediately crawls over and searches for it, confident that it still exists even though she can no longer see it.	
2	A 5-year-old insists that his tall, skinny cup has more juice in it than his friend's short, wide cup, even though both cups were filled with the exact same amount.	
3	An 8-year-old can correctly sort a mixed pile of buttons by color, then by size, then by shape, successfully switching between multiple logical categories.	
4	A 16-year-old debates with her friends about what society might look like if money didn't exist, confidently reasoning through a scenario she's never actually experienced.	

Word Bank

Hypothetical Thinking	*Fluid Intelligence*
Abstract Thinking	*Zone of Proximal Development (ZPD)*
More Knowledgeable Other (MKO)	*Dementia*
Scaffolding	*Concrete Operational Stage*
Formal Operational Stage	*Crystallized Intelligence*

3. Vygotsky's Sociocultural Alternative

While Piaget focused on biological maturation and independent discovery, Lev Vygotsky argued that children are social learners. He believed cognitive development is largely driven by cultural context and interactions with others.

Central to Vygotsky's theory is the role of a _____—anyone who has a better understanding or a higher ability level than the learner with respect to a particular task. While this is often an adult like a parent or teacher, an MKO can also be a more advanced peer or even a computer program!

Vygotsky proposed that learning happens best in the _____—the "sweet spot" between what a learner can do independently and what they can only achieve with guidance. To help a child move through this zone, the MKO provides _____. This is temporary, tailored support that is gradually removed as the child becomes more competent.

Example in Action: Imagine a child learning to ride a two-wheeled bike. Riding completely alone is currently beyond their ZPD, but riding with a parent (the MKO) holding the seat is within it. The parent holding the back of the seat acts as the scaffolding. As the child gains balance, the parent gradually loosens their grip, eventually removing the scaffolding until the child can pedal entirely on their own.

📺 *Video Notes: "The Growth of Knowledge" (Crash Course Psychology #18)*

Watch the video above, then answer each question below in your own words.

1. What does the video mean by maturation, and how did Piaget's observations of children's mistakes lead him to propose stages of cognitive development?

2. How does the video describe babies in the sensorimotor stage, and how does the blanket-and-pacifier example illustrate object permanence?

3. What is centration, how does the video connect it to conservation and reversibility in the preoperational stage, and what changes in the concrete operational stage?

4. How does the video illustrate egocentrism and theory of mind? Describe the example it uses for each.

5. What criticisms of Piaget's four-stage model does the video give, and what does it say was his greatest achievement despite them?

6. How does the video say Vygotsky's view of development differs from Piaget's?

Apply It: MKO, ZPD, or Scaffolding?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A chess coach who is a much stronger player than her student patiently points out mistakes and explains better strategies during their weekly lessons.	
2	A piano teacher gives a beginner student a simplified version of a song with color-coded keys, then gradually removes the color-coding as the student improves until she can read standard notes on her own.	
3	A student can't yet solve long division problems completely independently, but she can successfully work through them when her tutor is nearby offering occasional hints.	

4. Cognitive Changes in Adulthood

Cognitive development does not stop when childhood ends. As adults age, their intelligence shifts in different ways:

- _____: The ability to reason quickly and abstractly (solving novel logic problems) tends to peak in early adulthood and slowly wanes as people get older.
- _____: Our accumulated knowledge, vocabulary, and facts remain relatively stable and actually increase throughout late adulthood.

While normal aging brings minor memory decline, severe cognitive loss is not typical. _____ is a broad category of cognitive disorders (such as Alzheimer's disease) characterized by a severe and progressive decline in memory and daily functioning.

5. Don't Trip Up! (Common Misconceptions)

⚠️ Assimilation vs. Accommodation: These two terms are constantly tested and easily confused! Use this trick: Assimilation uses All the Same Schemas. Accommodation Creates Change.

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. A toddler who has an existing mental framework for "dog" sees a horse for the very first time and, without changing his understanding at all, confidently calls it a "big dog." This process of fitting new information into an existing mental framework is called:

- **A.** Object Permanence
- **B.** Assimilation
- **C.** Accommodation
- **D.** Scaffolding

2. A 4-year-old boy excitedly wraps up his own favorite toy truck as a birthday present for his mother, genuinely assuming she will love it exactly as much as he does, without considering that she might prefer something else entirely. This inability to see a situation from another person's point of view is called:

- **A.** Animism
- **B.** Conservation
- **C.** Egocentrism
- **D.** Reversibility

3. A researcher pours the exact same amount of juice into a short, wide glass and a tall, skinny glass in front of a 5-year-old child, and the child confidently insists the tall glass contains more juice simply because the liquid line appears higher. This child's mistaken belief demonstrates a lack of:

- **A.** Conservation
- **B.** Object Permanence
- **C.** Theory of Mind
- **D.** Reversibility

4. A 9-year-old girl can now easily understand that if 4 + 3 = 7, then 7 - 3 must equal 4, successfully performing this mental math operation involving concrete numbers. According to Piaget, which stage does this reasoning ability place her in?

- **A.** Formal Operational Stage
- **B.** Sensorimotor Stage
- **C.** Concrete Operational Stage
- **D.** Preoperational Stage

5. While a child is learning to read, her older brother, who reads at a much higher level, patiently helps her sound out difficult words and understand the story. According to Vygotsky, the older brother is functioning as a:

- **A.** Schema
- **B.** Reflex
- **C.** More Knowledgeable Other (MKO)
- **D.** Zone of Proximal Development

6. A child cannot yet ride a bike completely alone, but she can successfully ride it with her father holding onto the back of the seat for support and balance. According to Vygotsky, this "sweet spot" between what she can do independently and what she can only achieve with guidance is called the:

- **A.** Scaffolding
- **B.** Critical Period
- **C.** Sensorimotor Stage
- **D.** Zone of Proximal Development

7. As adults move into their later years, they generally maintain a large and stable vocabulary and continue accumulating general knowledge and facts, even as their ability to quickly solve brand-new, unfamiliar logic puzzles noticeably declines. Which type of intelligence remains stable or even increases throughout late adulthood?

- **A.** Crystallized Intelligence
- **B.** Fluid Intelligence
- **C.** General Intelligence (g)
- **D.** Formal Operational Thought

✅ Check Your Progress

Done with your notes? Head over to the Quiz to test what you've learned:
https://appsychreviewhub.com/quiz

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